

BACTERIA AND ARCHAEA**Activity 27.1 How diverse are the Archaea?**

1. The Archaea are subdivided into four major groups: the Korarcheotes, the Euryarcheotes, the Crenarcheotes, and the Nanoarcheotes.

a. What characteristics are used to place organisms into each of these groups?

Archean group	Characteristics
Korarcheotes	
Euryarcheotes	
Crenarcheotes	
Nanoarcheotes	

2. There is great diversity in the ways different species of microbes

- obtain energy for metabolic functions
- obtain carbon for building the macromolecules of life.

Fill in the chart to indicate how these characteristics are used to describe the nutritional type or nutritional classification of organisms. (Refer also to Table 27.1 in Campbell Biology, 11th edition.)

	Plants	Animals	Bacteria and/or Archaea			
Energy source	<i>Light</i>	<i>Organic compounds</i>	Light	Organic compounds	Light	Organic compounds
Carbon Source	<i>Inorganic CO₂</i>	<i>Organic compounds</i>				
Mode of Nutrition	<i>Autotroph</i>	<i>Heterotroph</i>				

3. Bacteria first appear in the fossil record about 3.5 billion years ago. Humans first appear in the fossil record only a few million years ago. Given this, which group would you say is more highly evolved?
 - a. What kinds of arguments or evidence would you use to support the idea that bacteria are more highly evolved?
 - b. How would you support the idea that humans are more highly evolved?
4. Given what you know about the origin of life on Earth, you want to look for life on other planets.
 - a. What characteristics of the planet's environment would you look for? Explain your reasoning.
 - b. What kind(s) of life would you look for? Explain your reasoning.
 - c. What tests or probes would you use to find the kind of life you proposed in part b? Explain your reasoning.

Activity 27.2 How has small size affected prokaryotic diversity?

1. All living organisms must maintain a relatively constant internal environment. Maintaining this environment means that a certain concentration of each substance must be maintained per unit volume of the cell. The ability of the cell to maintain a specific concentration of a substance is affected by

- the ability of the substance to diffuse through the membrane,
- the overall size and shape of the cell, and
- the maximum amount of the substance a given area of membrane can transport per unit time.

a. How do surface-area-to-volume (SA/V) ratios change as the size and shape of cells and organisms change? To answer this, calculate the SA and V of a cube 1 mm on a side. Then do the same for cubes that are 2 mm and 4 mm on a side and compare their SA/V ratios.

CUBES:	1 mm square	2 mm square	4 mm square
Linear dimension	1 mm	2 mm	4 mm
Surface area (SA)	$6 (1 \text{ mm}^2) = 6 \text{ mm}^2$		
Volume (V)	$(1 \text{ mm})^3$		
SA/V ratio	$6 \text{ mm}^2 / 1 \text{ mm}^3$		

b. In general, how does surface area change as linear dimensions increase twofold?

c. In general, how does volume change as linear dimensions increase twofold?

d. In general, how do SA/V ratios change as linear dimensions increase twofold?

2. Assume a bacterium is 10 mm in linear dimension. Fill in the chart.
- If modeled as a cube, what would its SA, V, and SA/V ratio be?
 - If modeled as a sphere, what would its SA, V, and SA/V ratio be?
 - What are the SA and V values and the SA/V ratios for a cube-shaped eukaryotic cell that is 100 mm in linear dimension?

	a. 10-mm bacterium as a cube	b. 10-mm bacterium as a sphere	c. 100-mm eukaryote, cube-shaped
SA			
V			
SA/V ratio			

3. Assume that every cell requires a minimum of 1 unit of oxygen per mm³ per second to stay alive. Fill in the chart.
- How much oxygen must cross each mm² of surface area per second in the 10-mm bacterium versus the 100-mm eukaryote to keep each alive?
 - What effects might this difference have on metabolic rates in these organisms?

	10 μm bacterium	100 μm eukaryote
a. Oxygen/μm ² of SA/second		
b. Possible effect(s) on metabolic rate		

4. Given what you know about cell membranes, is there likely to be a maximum upper limit on the number of molecules of a substance that can cross a given area of membrane per unit of time? If so, what factors would be involved in determining the maximum upper limit?

- 5.
- On average, how large is a prokaryotic genome?
 - On average, how many times larger is a eukaryotic genome?
 - Are these genomes haploid or diploid?

6. Under ideal conditions, many bacterial cells can reproduce or duplicate themselves within an hour. Some species (for example, *E. coli*) can reproduce every 20 minutes under ideal conditions. If you inoculate 1 liter of culture medium with one bacterium per milliliter (of medium) at time $t = 0$, how many bacteria will be present in 1 milliliter of the culture medium after 10 hours? (To do this calculation, assume that in these culture conditions the bacteria duplicate—or double in number—every hour.)

7. Assume that the mutation rate of bacterial cells in culture is 10^{-6} to 10^{-8} . This means that in every 1 million to 100 million cells produced from a single original cell, you would expect to find at least one mutation. How many mutations would you be likely to find in 1 liter of the 10-hour culture of the cells you grew in question 6?

8. Using all of the information in this activity, write an argument titled: “How small size in prokaryotes played a role in limiting their morphological diversity and promoting their biochemical (metabolic) diversity.”